Employee Management System

System Sequence Diagrams and Activity Diagrams

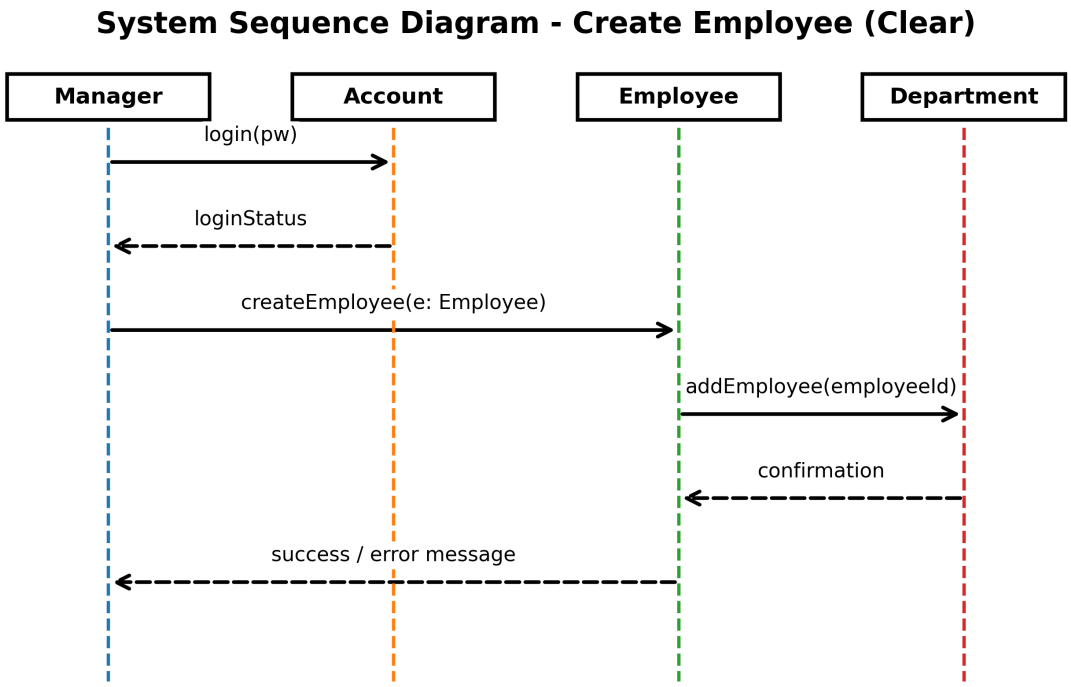
This document explains how the Employee Management System works.

Introduction

The Employee Management System helps manage employee information and attendance. This project includes system sequence diagrams and activity diagrams. The diagrams show how users interact with the system step by step.

1. System Sequence Diagram – Create Employee

This diagram shows how a manager creates a new employee in the system.

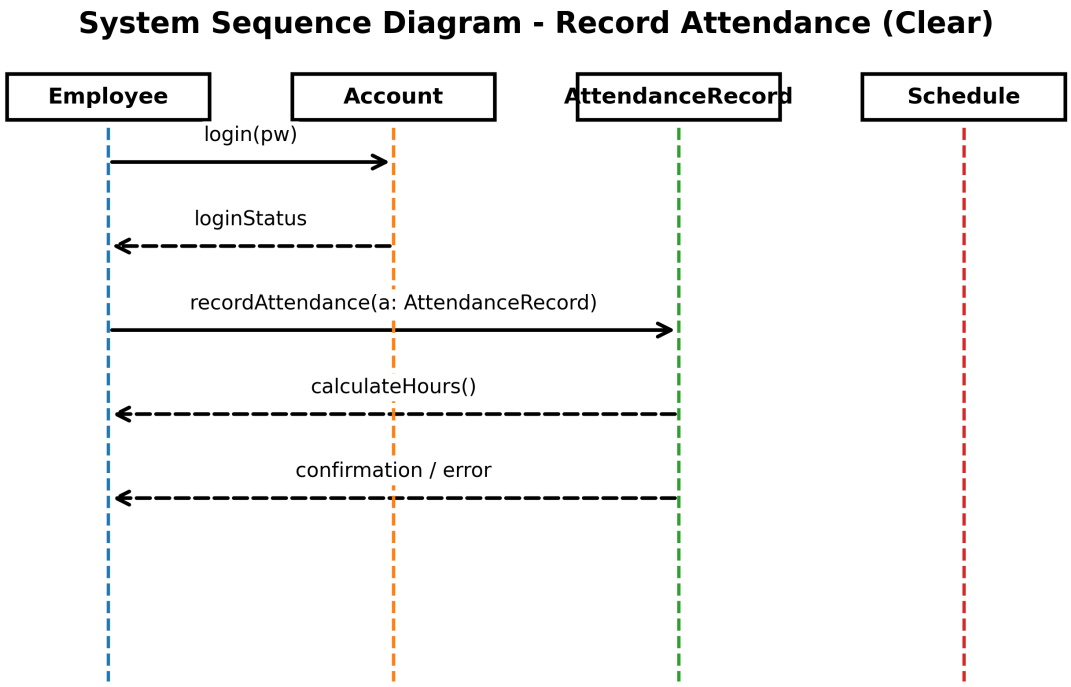


Explanation:

1. The manager logs into the system.
2. The Account class checks the login information.
3. The system returns the login status.
4. The manager creates a new employee.
5. The Employee class sends the employee ID to the Department.
6. The Department confirms the employee was added.
7. The system sends a success or error message back to the manager.

2. System Sequence Diagram – Record Attendance

This diagram shows how an employee records attendance.

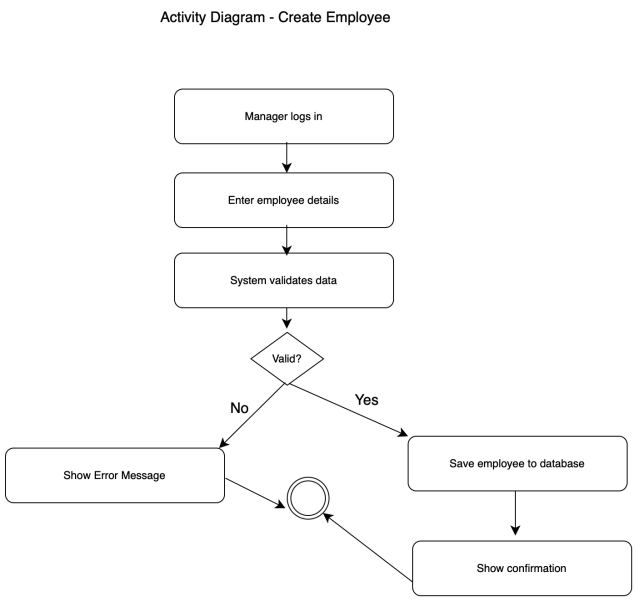


Explanation:

1. The employee logs into the system.
2. The Account class checks the login details.
3. The system sends back the login status.
4. The employee records attendance.
5. The Attendance Record class calculates hours worked.
6. The system sends confirmation or an error message.

3. Activity Diagram – Create Employee

This activity diagram explains the process of creating a new employee.

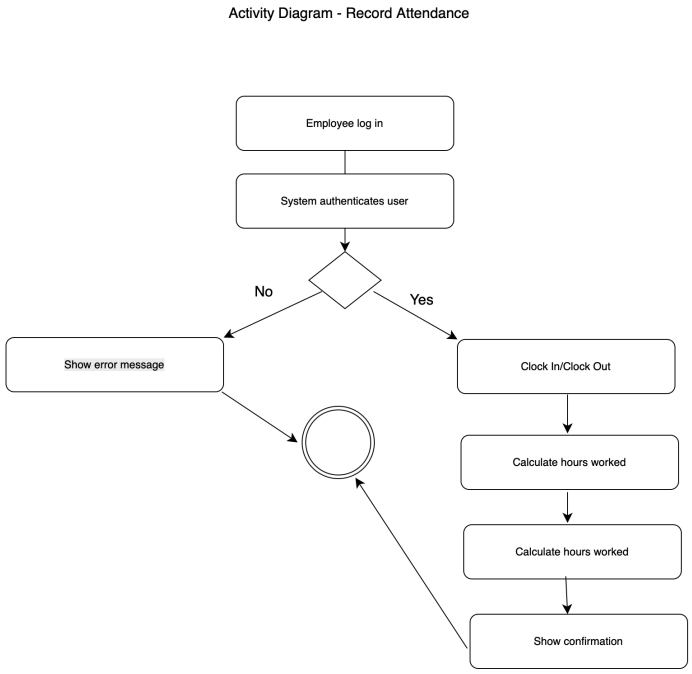


Explanation:

1. The manager logs into the system.
2. The manager enters employee details.
3. The system checks if the information is valid.
4. If valid, the system saves the employee to the database.
5. The system shows a confirmation message.
6. If not valid, the system shows an error message.

4. Activity Diagram – Record Attendance

This activity diagram explains how attendance is recorded.



Explanation:

1. The employee logs into the system.
2. The system checks if login is successful.
3. If login works, the employee clocks in or clocks out.
4. The system calculates the hours worked.
5. The system updates the attendance record.
6. The system shows confirmation.
7. If login fails, an error message is shown.

Conclusion

The Employee Management System allows managers to create employees and employees to record attendance. The system sequence diagrams show how classes communicate with each other. The activity diagrams show the each steps process. Together, these diagrams help explain how the system works clearly.